

# Karan Shah

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## Education

### Stony Brook University

Stony Brook, NY

MS IN COMPUTER SCIENCE | GPA : 3.68/4

Aug. 2019 - Exp. Dec. 2020

Coursework: Theory of Database Systems, Big Data Analytics, Machine Learning, Computer Vision, Data Science, Probability and Statistics

### Gujarat Technological University

Ahmedabad, India

B.E. IN COMPUTER ENGINEERING | GPA : 8.03/10

Aug. 2014 - May 2018

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

## Work Experience

### Software Developer Intern

Sept. 2020 - Current

QUANTIPHI

Marlborough, MA

- Building software tools in **Python and React.js** with Machine Learning Engineers to accelerate production-grade Conversational AI use cases in **Docker Container** over **GCP** cluster
- Using **Flask** based routing methods with **multiprocessing** to trigger underlying capabilities of **NVIDIA's** end-to-end ML framework and **Socket+GRPC** for live audio transcription

### Software Engineer

June 2018 - April 2019

SIMFORM

Ahmedabad, India

- Developed an end to end **Video Streaming system**; allows user to upload and scan the content on the cloud, stream on cross-platform desktop application over globally separated clients; using **AWS services (Lambda, S3, Cloudfront, MQTT)**, **Electron.js**, and **React.js**
- Built a software system to semi-automate Company's marketing work with Chrome Extension and **REST** API Web backend using **Python and Node.js**, efficient in reducing ~ **40 %** of Marketing team's manual work
- Designed and automated **ETL** jobs to process a huge realtime data using AWS services (Glue, Lambda, S3) and **Spark**

### Research Intern, Machine Learning

July 2017 - May 2018

ISRO - INDIAN SPACE RESEARCH ORGANIZATION

Ahmedabad, India

- Designed generalized **CNN** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for preprocessing of geographic remote sensing data, visualization, real-time performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

## Skills

**Programming** Python, JAVA, Javascript, C/C++, Matlab

**Cloud/Big Data** AWS, Google Cloud Platform, MapReduce, Spark

**Tools/Technologies** Docker, Kubernetes, Git, REST, GRPC, SQL, NoSQL

**Frameworks/Libraries** Node.js, Django, Flask, React.js, PyTorch, Tensorflow, Keras, OpenCV

## Publication

### Band-wise Independent Pansharpening Using Neural Networks with Shared Weights

PANKAJ BODANI, KARAN SHAH, SHASHIKANT SHARMA (ISRO)

SUBMITTED TO **GEOCATO INTERNATIONAL JOURNAL**

## Projects

### Medical Search Engine, RadSearch (Java, Python, Javascript, Elastic Search)

Jan. 2020 - May 2020

- Created a distributed, multitenant-capable full-text search engine with an HTTP web interface and schema-free JSON documents using **Elastic Search** to handle user queries and **Java Servlets, Python, Javascript** to handle the backend
- Currently being used regularly by the doctors at the Stony Brook Hospital to find the most relevant information about patients

### Company's Internal Communication System (Java, Javascript)

Sept. 2018 - Dec. 2018

- Created a web-based software for employees of a Software Company to send/receive messages, share files in person, groups and broadcast channels with RBAC - Roll based access control
- Utilised **SQLite** to store data of each user with **Java** Servlets to query the database and applied Software Engineering approaches

### Retail Sales Data Analysis (Python, Pandas)

Oct. 2019 - Dec. 2019

- Analyzed the market basket dataset to provide sales insight for a chain of retail stores.
- Identified reasons for sales drop with the help of external datasets
- Implemented **Word2Vec** and **LSTM model with k-means clustering** to improve Shelf Space Management and Inventory management

### Video Action Classification - Recognition (Python, PyTorch, Matlab)

Nov. 2019 - Dec. 2019

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (1st in class) on image test data, and **84%** (2nd in class) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

### Desktop Application to Synchronize Local Storage with Cloud (AWS, Node.js, Electron.js)

March 2017 - Apr. 2017

- Built cross-platform application leveraging the Electron.js framework
- Used **Node.js** and **AWS services** to fully synchronize local storage and AWS S3 storage